DS	**Session**	Sorting	**Question Information**	Level 1 Challenge 17

Question Description:

Simon is studying B.Tech.-Mechanical Engineering.

He's going to attend a computer science-based subject exam this semester.

Due to the less preparation in the previous monthly tests, his internal mark decreased.

His computer science Professor made an offer one more chance to boost up his internal marks.

Professor assigns a program to Simon for the internal mark bootup.

So Simon wants to solve the given task is, Given two arrays, A and B, of equal size n,

the task is to find the minimum value of A[0] * B[0] + A[1] * B[1] +...+ A[n-1] * B[n-1],

where shuffling of elements of arrays A and B is allowed.

can you help him in solving Questions ?

Constraints:

1<=T<=100

1<=N<=50

1<=A[]<=20

Input Format:

The first line of input contains an integer denoting the no of test cases.

Then T test cases follow. Each test case contains three lines.

The first line of input contains an integer N denoting the size of the arrays.

In the second line are N space separated values of the array A[], and in the last line are N space separated values of the array B[].

Output Format:

For each test case in a new line print the required result.

Example :

Input : A[] = {3, 1, 1} and B[] = {6, 5, 4}.

Output : 23 Minimum value of S = 1*6 + 1*5 + 3*4 = 23.

Input : A[] = { 6, 1, 9, 5, 4 } and B[] = { 3, 4, 8, 2, 4 }

Output : 80. Minimum value of S = 1*8 + 4*4 + 5*4 + 6*3 + 9*2 = 80.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3
4
2 5 3 7
8 5 7 3
3
13 11 1
6 15 14
5
6 11 9 15 4
3 41 8 21 4
```

Test Case 2

INPUT (STDIN)

```
2
3
13 11 1
6 15 14
5
6 11 9 15 4
3 41 8 21 4
```

EXPECTED OUTPUT

```
247
451
```

EXPECTED OUTPUT

```
83
247
451
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void sort(int a[],int n,int
flag)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
1
```

Test Case 2

TOKEN COUNT

```
420
```

Test Case 3

NLOC

```
55
```

Code

solution.c

```c
#include <stdio.h>

void sort(int a[], int n, int flag) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = i + 1; j < n; j++) {
            if ((flag == 1 && a[i] > a[j]) || (flag == 0 && a[i] < a[j])) {
                int t = a[i]; a[i] = a[j]; a[j] = t;
            }
        }
    }
}

int main() {
    int t;
    if (scanf("%d", &t) != 1) return 0;
    while (t--) {
        int n;
        scanf("%d", &n);
        int a[n], b[n];
        for (int i = 0; i < n; i++) scanf("%d", &a[i]);
        for (int i = 0; i < n; i++) scanf("%d", &b[i]);
        sort(a, n, 1);
        sort(b, n, 0);
        long long sum = 0;
        for (int i = 0; i < n; i++) sum += (long long)a[i] * b[i];
        printf("%lld\n", sum);
    }
    return 0;
}
```

Output

Problems Output Debug Console Terminal Ports powershell

```
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
2
3
13 11 1
6 15 14
5
6 11 9 15 4
3 41 8 21 4
247
451
PS C:\C Programming>
```